Hanwha Vision (formerly Samsung Techwin / Hanwha Techwin)

Hanwha Vision (formerly Samsung Techwin / Hanwha Techwin) — Firmware Attack Surface & Detection Brief

Scope: CVE and public-advisory (OSINT) research only — **no firmware or software has been acquired or extracted for this vendor.** Every finding below comes from public vulnerability disclosures (NVD, CISA ICS advisories, vendor PSIRT) and product documentation. CVE IDs listed in the Confirmed table were each independently re-verified against NVD/CISA/vendor records before inclusion; candidate IDs that could not be confirmed were dropped, not guessed.

Identity

Field	Value
Census / OUI string	`Hanwha Vision VietNam`
Operating company	Hanwha Vision Co., Ltd. (Seongnam, South Korea), a subsidiary of Hanwha Group. "Hanwha Vision VietNam" is the company's Vietnam manufacturing subsidiary (60,000 sq m smart factory in Que Vo Industrial Park, Bac Ninh Province, opened 2017, mass production from 2018). The Vietnam plant is where Wisenet cameras/NVRs/lenses are built and MAC addresses are burned, so the OUI/registered name resolves to that entity, but the product line, firmware, and vulnerability response all belong to Hanwha Vision Co., Ltd.
OT devices (ARNG census)	72 across 4 states
Genuine OT vendor?	Yes
Purdue placement	Levels 0-2 of a physical-security OT network. Cameras are edge sensors/field devices (L0 image sensor / L1 embedded device); NVRs doing local recording and the VMS doing local viewing sit at L2 (local supervision). Management tooling - Wisenet Device Manager and WAVE VMS servers/consoles - sits at L2-L3 (supervisory/operations), and on converged installs the security VLAN typically crosses into the IT/DMZ at L3-L3.5. Justification: cameras generate/forward streams but are managed, so they are edge devices; the NVR/VMS is the local control-and-record layer; WDM is the human-management layer.

Identity note. Census string "Hanwha Vision VietNam" is an OUI/registered-name artifact of the manufacturing subsidiary, NOT a different company. Corporate lineage: Samsung Aerospace/Precision (late 1970s) -> Samsung Techwin -> acquired by Hanwha Group 2015 and renamed Hanwha Techwin -> rebranded Hanwha Vision in March 2023. The 72 census devices are genuine Hanwha/Samsung-Techwin physical-security products (Wisenet IP cameras, NVRs). Asset-owners should expect the installed base to span three brand generations of device labels/firmware (Samsung Techwin, Hanwha Techwin, Hanwha Vision) that are the

same product family. Note: some OUIs on these devices may also appear registered to "Hanwha Techwin" or legacy "Samsung Techwin" ranges - treat all three as the same vendor for inventory.

Product lines & OT relevance

Core physical-security OT: IP video surveillance (CCTV) and video management, deployed as force-protection/perimeter monitoring on National Guard facilities. These devices routinely bridge the physical-security VLAN to IT and to the facility environment, making them a recurring IT/OT pivot and a high-value target for both surveillance-blinding and lateral movement.

- Wisenet network/IP cameras - fixed, dome, bullet, PTZ, multi-sensor, AI, thermal (e.g. QNV-C8012, QND-8080R, and the P/Q/X/T/L series)
- Network Video Recorders (NVRs) and legacy DVRs (Wisenet recorders)
- WAVE VMS and SSM/SmartViewer video management software
- Wisenet Device Manager (WDM) - Windows fleet discovery/management/firmware tool
- WAVE/SolidEDGE SSD-based edge-storage cameras running VMS at the edge
- Thermal imaging and radiometric cameras
- Camera lenses, storage/accessory devices, and integrated access/security peripherals

Attack surface

- HTTP/HTTPS web management UI with proprietary CGI endpoints (e.g. /stw-cgi/*, /cgi-bin/stw.cgi) - source of command injection and XSS flaws
- ONVIF (TCP 80/443) plus WS-Discovery device discovery (UDP 3702)
- RTSP video streaming (TCP 554), with RTP media ports
- Legacy default credentials: older Samsung/Hanwha Techwin cameras historically shipped with weak admin defaults (e.g. admin/4321); Wisenet Device Manager held hard-coded/shared default credentials (CVE-2025-52601). Modern firmware forces password-set on first boot, but unmanaged legacy units may still carry defaults
- ThroughTek Kalay P2P cloud connectivity embedded in NVRs - outbound P2P tunnels to vendor cloud (CVE-2021-28372)
- Wisenet Device Manager (Windows app) storing device credentials in UserConfig.xml readable by all local Windows users
- FTP/SMTP event delivery used to push analytics reports (abused in CVE-2025-52600)
- Unauthenticated/auth-bypass and OS command injection in the camera web server (CVE-2024-54012 / 54013)
- Firmware update channel via web UI and WDM; legacy devices may expose Telnet/SSH/SNMP/UPnP/mDNS

Confirmed CVEs / advisories

Each row was re-verified against an authoritative record (URL given). This is the public disclosure history, not an exhaustive list.

CVE / advisory	Applies to	Summary	Source
`CVE-2024-54013`	Named-product CVE for Hanwha Vision network cameras (QND-8080R and a long list of other Ha	Request-handling flaw in the Hanwha Vision camera web server allows authentication bypass / access to protected functions under certain conditions (CWE-306 miss	https://nvd.nist.gov/vuln/detail/CVE-2024-54013
`CVE-2024-54012`	Named-product CVE for Hanwha Vision network cameras (QND-8080R plus many other models). Co	OS command injection via improper input validation in the Hanwha Vision camera web server (CWE-78); specially crafted requests run malicious commands on the dev	https://nvd.nist.gov/vuln/detail/CVE-2024-54012
`CVE-2025-52601`	Affects Wisenet Device Manager, Hanwha's Windows fleet-management tool used to administer	Hard-coded cryptographic key in Hanwha Vision Device Manager (WDM) lets an attacker decrypt stored sensitive information, including shared/default device creden	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2025-52601
`CVE-2025-52600`	Named-product CVE spanning a wide range of Hanwha Vision network cameras. The malicious re	Client-side enforcement of server-side security / improper input validation in Hanwha Vision camera video-analytics (People Counting) reporting (CWE-602/CWE-20)	https://nvd.nist.gov/vuln/detail/CVE-2025-52600
`CVE-2021-28372`	Supply-chain CVE: Hanwha NVRs embed the ThroughTek Kalay P2P SDK for cloud/remote connecti	Improper access control / device impersonation in ThroughTek's Kalay P2P SDK (Kalay Platform 2.0); an attacker with a device UID can hijack a victim's P2P conne	https://nvd.nist.gov/vuln/detail/CVE-2021-28372

Detection guidance

- Inventory by OUI across Hanwha Vision / Hanwha Techwin / Samsung Techwin ranges; reconcile against the 72 census devices and flag any beyond the 4 expected states
- Monitor and baseline RTSP (TCP 554), HTTP/HTTPS (TCP 80/443), and ONVIF WS-Discovery (UDP 3702); alert on new talkers on these ports in the security VLAN
- Allowlist camera egress: a camera/NVR should only reach its NVR/VMS, NTP, DNS, and the firmware host. Alert on ANY camera or NVR initiating outbound Internet connections
- Specifically detect/deny outbound ThroughTek Kalay/TUTK P2P traffic (UDP to Kalay cloud) from NVRs - P2P should be disabled in an OT enclave
- Alert on HTTP access to proprietary CGI paths (/stw-cgi/eventsources.cgi, /cgi-bin/stw.cgi) from any host that is not the authorized VMS/WDM management station

- Collect logs from NVR/VMS audit trails, camera syslog, WDM activity, and the firewall at the security-VLAN boundary; watch WDM host filesystem for reads of UserConfig.xml
- Flag devices running firmware below patched baselines (camera X-series fixed in 2.24.00; WDM fixed in 2.9.3.1) as part of vuln management
- Segment the physical-security VLAN from IT/OT; force all management through a dedicated jump host and alert on management traffic originating elsewhere
- Watch FTP/SMTP event-delivery destinations configured on cameras for unexpected external targets

Acquisition status

We currently hold NONE for this vendor. Firmware and software are distributed from Hanwha Vision's support portals (hanwhavision.com and hanwhavisionamerica.com) and are pulled/applied through the free Wisenet Device Manager (WDM) tool and WAVE/SSM VMS installers. Management software (WDM, WAVE, SmartViewer) is generally freely downloadable, while per-model camera/NVR firmware images are typically gated behind a free registered support account (partner/dealer tiers get broader access); some firmware is only delivered in-band via WDM rather than as a standalone file. S-CERT vulnerability reports are public PDFs on the vendor site. Realistic acquisition paths: (1) register a free support account and download management tooling plus available firmware; (2) obtain firmware in-band from a lab device via WDM; (3) extract firmware from a physical unit. Acquisition is therefore partially gated - low friction for tooling, account-gated for many firmware images.

Notes

Verification discipline: every CVE above was confirmed in THIS research by fetching a live authoritative source - the NVD REST API (services.nvd.nist.gov) for each ID and the CISA page (cisa.gov/news-events/ics-advisories/icsa-21-229-01) for the ThroughTek advisory; the two Nozomi-disclosed 2025 IDs were cross-checked against the Nozomi Networks Labs write-up. No CVE here is from memory. Additional candidate IDs surfaced in search but were NOT individually verified and are therefore excluded from cve_claims: CVE-2025-52598/52599/8075 (Nozomi set - cert-validation, permission, XSS), CVE-2024-54011, CVE-2023-31994/31995/31996, CVE-2023-5037/5038, CVE-2021-32934, and CVE-2018-6301. Hanwha publishes its own S-CERT vulnerability-report PDFs (hanwhavision.com/.../wp-content/uploads/...) which are the authoritative vendor advisory source for follow-up. No standalone CISA/ICS-CERT advisory dedicated to Hanwha-branded cameras was found in this pass (only the upstream ThroughTek advisory). We hold no firmware/software, so this brief is OSINT/advisory only; detection guidance is behavior/protocol-based and does not depend on hash or firmware acquisition.

Status

Item	State
Firmware/software acquired	**No** — OSINT/advisory only
Confirmed CVEs	5 (independently re-verified)
Behavioural baseline	None — no samples held
Next step	CISA/PSIRT deep pass; attempt firmware acquisition

Generated 2026-10-26 from the ot-vendor-briefs workflow (12 researchers + adversarial CVE verification). Device counts from the 2026-10-22 ARNG OT census.